Digital Access & Economic Productivity: Cross-Country Data Analysis

Metrics & Data Sources

Metrics

- **Digital Access:** Individuals using the Internet (% of population)
- **Economic Productivity:** GDP per person employed, PPP-adjusted, constant prices

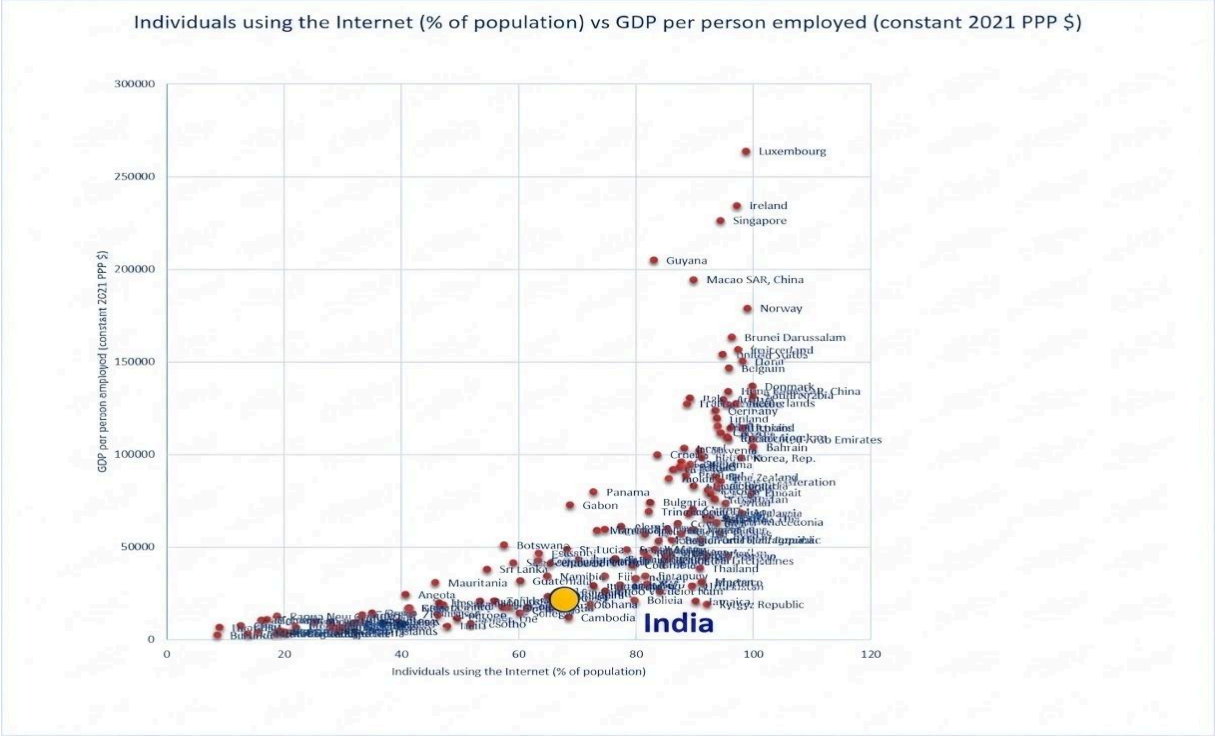
Data Sources

- **Internet Usage:** World Bank World Development Indicators (WDI)
- **Economic Productivity:** World Bank World Development Indicators (WDI)

Data Segmentation: Segment the data: By World Bank income group:

- **High-income countries:** Luxembourg, Singapore, Ireland, Switzerland, USA, Germany, UK, Japan, South Korea, France, Qatar, Belgium, Denmark, Netherlands, Australia
- **Upper-middle-income countries:** China, Brazil, Vietnam, Indonesia, South Africa, Mexico, Russia, Guyana, Macao SAR, Brunei, Saudi Arabia, UAE, Panama, Costa Rica, Malaysia, Kazakhstan, Turkey, Bulgaria
- **Lower-middle-income countries:** India, Pakistan, Bangladesh, Nigeria, Kenya, Egypt, Ghana, Philippines, Sri Lanka, Morocco, Ukraine, Uzbekistan, El Salvador, Bolivia
- **Low-income countries:** Ethiopia, Uganda, Rwanda, Malawi, Mozambique, Burundi, Chad, Madagascar, Afghanistan, Niger, Sierra Leone, Haiti.

SCATTER PLOT



Findings

High internet penetration is correlated with higher average productivity, as the dataset shows a clear upward progression across internet usage tiers.

1. Low Internet Access Acts as a Floor on Productivity

In the scatter plot, every country with less than 40% internet adoption remains tightly clustered at the bottom of the chart with a GDP per person employed under $20,000 (such as Burundi, Chad, Uganda, Mozambique, Ethiopia, Afghanistan, and Niger). The data shows no instance of a country achieving high productivity without first achieving significant internet penetration.

2. High Internet Access Does Not Guarantee High Productivity

The data shows widespread variance in productivity among countries with similar high internet access rates:

- **Luxembourg / Ireland vs. Vietnam:** Luxembourg has internet penetration above 95% with a GDP per worker of ~$263,700, and Ireland reaches ~$234,300, whereas Vietnam has an internet penetration of 84.15% with a GDP per worker of $26,057—showing nearly a 10-fold difference despite comparable high digital coverage.
- **China vs. Brazil vs. Mexico vs. Emerging Tiers:** China (92.00% internet, $45,834 GDP/worker), Brazil (84.46% internet, $41,441 GDP/worker), Mexico (83.12% internet, $48,518 GDP/worker), and

Thailand (~85% internet, ~$35,000 GDP/worker) have achieved digital reach similar to advanced economies, yet their productivity remains in the mid-tier.

- **United States & Advanced Economies vs. Russia:** The United States stands at $153,969 GDP per worker with 94.69% internet penetration, significantly outperforming countries like France ($127,332), Germany ($123,669), and the UK ($109,200), while Russia sits at $85,392 despite having high internet adoption (94.36%).
- **Resource & Specialized Hub Outliers:** High capital and resource-heavy economies like Guyana ($204,800 GDP/worker at ~78% internet), Macao SAR ($194,100), Norway ($178,800), and Brunei Darussalam ($163,200) achieve extreme productivity outcomes driven by sector-specific dynamics rather than digital infrastructure alone.
- **India and its neighbours:** India appears in the **lower-middle portion of the graph**: its internet penetration is substantially higher than many poorer countries, but GDP per person employed remains far below the high-income cluster.

  India has moved relatively far along the digital-access dimension, but has not moved an equivalent distance along the productivity dimension. In other words, India's position suggests that **having a large share of the population connected to the internet does not automatically result in high productivity per employed person.**

  - India has higher internet penetration than Bangladesh, Nepal, Pakistan and Sri Lanka in the 2024 data.
  - India is well below China and Bhutan in internet penetration.
  - India's GDP per person employed is above Bangladesh, Nepal, Pakistan and Myanmar, but below Bhutan, Sri Lanka and China.
  - Within the current lower-middle-income group, India's 2024 internet penetration (~65%) is above the group average (~55%), and its productivity (~$23,430) is also above the group's average (~$21,167).

Impact Assessment Based on the Data

The data indicates that internet access is a necessary baseline component for high economic productivity, but not the sole driver. While expanding internet access moves economies out of the lowest productivity bracket (<$20,000), reaching the highest productivity brackets ($100,000–$250,000+) depends on structural economic factors, industry composition, resource endowment, and capital depth beyond internet coverage alone.